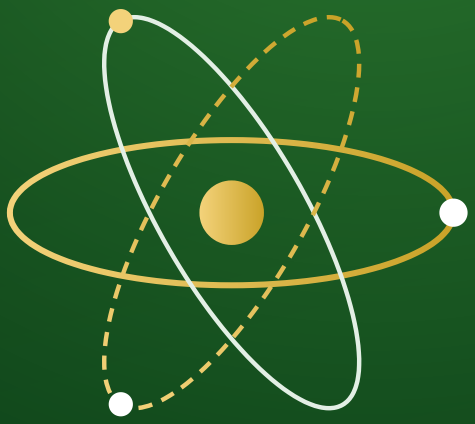
Freshman A+

ETHIOPIAN UNIVERSITY FRESHMAN EXAM PREP

LESSON-BY-LESSON NOTES

General Physics



clear concepts · solved problems

PHYS 1011 · CHAPTERS 1 – 7

THINK LIKE A PHYSICIST

$F = ma$	NEWTON II
$v^2 = v_0^2 + 2a\Delta x$	KINEMATICS
$P + \frac{1}{2}\rho v^2 + \rho gh = const$	BERNOULLI
$\Delta U = Q - W$	1ST LAW
$V = IR$	OHM

1. **Every concept, made clear**
   Simple explanations, diagrams and key formulas with units
2. **Solved step by step**
   Worked examples showing given → formula → substitution → answer
3. **Real university exams**
   AAU · AASTU · ASTU · Arba Minch · Bahir Dar · Hawassa and more, all with full solutions

01 Preliminaries
02 Kinematics & Dynamics
03 Fluid Mechanics
04 Heat & Thermo-dynamics
05 Oscillations, Waves & Optics
06 Electro-magnetism & Electronics
07 Physics in Science & Technology

Based on the MoSHE **General Physics** module

References: Serway (6th ed.) · OpenStax *University Physics* 1–3

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Seven chapters · lesson-by-lesson notes, worked examples, exam traps, and university exam questions with full solutions.

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HOW EACH LESSON IS BUILT

KEY IDEA
The main point you must remember.

EXAM TRAP
Where students usually lose marks.

MEMORY AID
A short trick to recall lists and pairs.

WORKED EXAMPLE
A problem solved step by step.

Question tags: University · Exam real past paper Serway / OpenStax reference book University · Model Exam prepared model question

CHAPTER

01

Preliminaries

The word **physics** comes from the Greek word for *nature*. Physics studies matter and energy, and it is above all **a science of measurement**: every law is tested by measuring something. This chapter gives the tools you need for the whole course: **physical quantities and SI units**, **uncertainty and significant figures**, and **vectors**. Almost every later problem uses these skills, and they are tested in every mid exam.

LESSONS IN THIS CHAPTER

1.1 Physical Quantities, Units and Dimensions

1.2 Uncertainty in Measurement and Significant Figures

1.3 Vectors: Composition and Resolution

Freshman A+ · General Physics (Phys 1011)

CHAPTER 1 · LESSON 1.1

Physical Quantities, Units and Dimensions

AFTER THIS LESSON YOU CAN

- Distinguish basic and derived quantities
- Name the 7 SI base quantities and units
- Use metric prefixes and convert units
- Use dimensions to check equations

1.1.1 Physical quantities and measurement

DEFINITIONS

- **Physical quantity**: a property of a body or phenomenon that can be **measured and expressed as a number with a unit** (the length of a rod, the mass of a body).
- **Measurement**: the act of **comparing** a physical quantity with a known **standard**.
- **Unit**: the standard quantity used for comparison (metre, kilogram, second...).

Every measured value has **two parts**: a **number** and a **unit**. "The table is 2" means nothing; "the table is 2 m long" is a measurement.

Type	Meaning	Examples
Basic (fundamental) quantities	Cannot be expressed in terms of other quantities	length, mass, time, temperature, electric current, amount of substance, luminous intensity
Derived quantities	Expressed as **combinations** of basic quantities	area, volume, density, velocity, force, pressure, energy, power

ALL OF PHYSICS FROM SEVEN QUANTITIES

Every physical quantity, however complicated, can be expressed using only **seven basic quantities**. That is what makes the SI system so powerful.

1.1.2 The SI system

The **SI** (Système International, International System of Units) is the modern metric system, adopted in **1960** at the 11th General Conference on Weights and Measures. It is built on **7 base quantities**:

Base quantity	Symbol	SI unit	Unit symbol	Dimension
Length	l, x	metre	m	[L]
Mass	m	kilogram	kg	[M]
Time	t	second	s	[T]
Electric current	I	ampere	A	[I]
Thermodynamic temperature	T	kelvin	K	[Θ]

Base quantity	Symbol	SI unit	Unit symbol	Dimension
Amount of substance	n	mole	mol	[N]
Luminous intensity	I_v	candela	cd	[J]

M **"MY LITTLE TINY CAT TAKES A LOLLIPOP"**

Mass (kg) · **L**ength (m) · **T**ime (s) · **C**urrent (A) · **T**emperature (K) · **A**mount of substance (mol) · **L**uminous intensity (cd).

Common derived quantities

Quantity	Definition	SI unit	In base units	Dimension
Area	length × width	m^2	m^2	$[L^2]$
Volume	l × w × h	m^3	m^3	$[L^3]$
Density	mass / volume	kg/m^3	$kg \cdot m^{-3}$	$[ML^{-3}]$
Velocity	displacement / time	m/s	$m \cdot s^{-1}$	$[LT^{-1}]$
Acceleration	velocity / time	m/s^2	$m \cdot s^{-2}$	$[LT^{-2}]$
Force	mass × acceleration	newton (N)	$kg \cdot m \cdot s^{-2}$	$[MLT^{-2}]$
Pressure	force / area	pascal (Pa)	$kg \cdot m^{-1} \cdot s^{-2}$	$[ML^{-1}T^{-2}]$
Work, energy	force × distance	joule (J)	$kg \cdot m^2 \cdot s^{-2}$	$[ML^2T^{-2}]$
Power	work / time	watt (W)	$kg \cdot m^2 \cdot s^{-3}$	$[ML^2T^{-3}]$
Momentum	mass × velocity	kg·m/s	$kg \cdot m \cdot s^{-1}$	$[MLT^{-1}]$
Electric charge	current × time	coulomb (C)	A·s	[IT]

! **UNITS THAT LOOK ALIKE**

- **N/m^2** is the unit of **pressure** (pascal), not density. Density is **kg/m^3** (or g/cm^3).
- **Amount of substance (mole)** is a **basic** quantity, even though it sounds derived.
- **Force is a derived** quantity (Arsi exam: "force is a basic quantity" → **False**).

Metric prefixes

Prefix	Symbol	Factor	Prefix	Symbol	Factor
tera	T	10^{12}	deci	d	10^{-1}
giga	G	10^{9}	centi	c	10^{-2}
mega	M	10^{6}	milli	m	10^{-3}
kilo	k	10^{3}	micro	μ	10^{-6}
hecto	h	10^{2}	nano	n	10^{-9}

Prefix	Symbol	Factor	Prefix	Symbol	Factor
deka	da	10^{1}	pico	p	10^{-12}

1.1.3 Conversion of units

Method: multiply by **conversion factors** (fractions equal to 1) arranged so that the unwanted unit **cancels**.

USEFUL CONVERSION FACTORS

1 in = 2.54 cm = 0.0254 m · 1 ft = 0.3048 m · 1 mile = 1.609 km · 1 lb = 0.4536 kg · 1 h = 3600 s · 1 L = 1000 cm³ = 10^{-3} m³

Temperature: $T_C = \frac{5}{9}(T_F - 32),\ T_K = T_C + 273.15$

WORKED EXAMPLE 1: SPEED CONVERSION

Convert 100 km/h to m/s.

$$100\ \tfrac{\text{km}}{\text{h}} \times \frac{1000\ \text{m}}{1\ \text{km}} \times \frac{1\ \text{h}}{3600\ \text{s}} = \frac{100\,000}{3600}\ \text{m/s} = 27.8\ \text{m/s}$$

Shortcut: to go from km/h to m/s, divide by **3.6**; from m/s to km/h, multiply by 3.6.

WORKED EXAMPLE 2: VOLUME CONVERSION

How many cubic metres are in 250 000 cm³?

Since $1\ \text{cm} = 10^{-2}\ \text{m}$, we get $1\ \text{cm}^3 = (10^{-2})^3\ \text{m}^3 = 10^{-6}\ \text{m}^3$.

$$250\,000\ \text{cm}^3 \times 10^{-6}\ \tfrac{\text{m}^3}{\text{cm}^3} = 0.25\ \text{m}^3$$

CUBE THE FACTOR FOR VOLUMES, SQUARE IT FOR AREAS

$1\ \text{m} = 100\ \text{cm}$, but $1\ \text{m}^2 = 10^4\ \text{cm}^2$ and $1\ \text{m}^3 = 10^6\ \text{cm}^3$. Forgetting to square or cube the factor is one of the most common exam mistakes.

1.1.4 Dimensions and dimensional analysis

The **dimension** of a quantity shows how it is built from the base quantities, written with square brackets: $[v] = [\text{L}][\text{T}]^{-1} = \text{LT}^{-1}$.

★ RULES OF DIMENSIONAL ANALYSIS

1. You can only **add, subtract or equate** quantities with the **same dimensions** (you cannot add a length to a time).
2. Both sides of a correct equation must have the **same dimensions** (the equation is *dimensionally consistent*).
3. Arguments of functions such as $\sin$, $\cos$, e^x and $\log$ must be **dimensionless**.
4. Pure numbers (2, $\frac{1}{2}$, π) are dimensionless, so dimensional analysis **cannot find numerical constants**.

✎ WORKED EXAMPLE 3: CHECKING AN EQUATION

Is $x = v_0 t + \frac{1}{2}at^2$ dimensionally correct?

$[x] = \mathrm{L} \cdot [v_0 t] = \mathrm{LT}^{-1} \cdot \mathrm{T} = \mathrm{L} \cdot [at^2] = \mathrm{LT}^{-2} \cdot \mathrm{T}^2 = \mathrm{L}.$

Every term has dimension L, so the equation is **dimensionally consistent**. (This does not prove it is right; the ½ could be wrong, but no dimensional error exists.)

✎ WORKED EXAMPLE 4: FINDING THE DIMENSIONS OF CONSTANTS (HAWASSA TEST)

In $P = \dfrac{b - x^2}{a\,t}$, P is power, x is distance and t is time. Find the dimensions of a and b.

Step 1: b is subtracted from x^2, so it must have the same dimension: $[b] = \mathrm{L}^2$.

Step 2: $[P] = \dfrac{[b]}{[a][t]} \Rightarrow [a] = \dfrac{\mathrm{L}^2}{\mathrm{ML}^2\mathrm{T}^{-3} \cdot \mathrm{T}} = \mathrm{M}^{-1}\mathrm{T}^2.$

Answer: $[a] = \mathrm{M}^{-1}\mathrm{L}^0\mathrm{T}^2$, $[b] = \mathrm{M}^0\mathrm{L}^2\mathrm{T}^0$.

Exam Practice

10 QUESTIONS · LESSON 1.1

1 Arba Minch University · Mid Exam

Which one of the following is NOT a unit of density?

A) kg/m³ **B)** g/cm³
C) N/m² **D)** g/mL

2 Arba Minch University · Mid Exam

Which of the following is a basic (fundamental) quantity rather than a derived one?

A) Area **B)** Power
C) Amount of substance **D)** Density

3 Arsi University · Mid Exam 2014

Force is a basic physical quantity.

True / False

4 Arba Minch University · Mid Exam

Which of the following lengths is the smallest?

A) 3.54×10^{-3} μm
B) 3.54×10^{5} mm
C) 3.54×10^{-3} mm
D) 3.54×10^{4} μm

5 Adama Science and Technology University · Mid Exam

Convert 100 km/h to m/s and write the dimension of velocity in terms of fundamental quantities.

6 Hawassa University · Test

In $P = \dfrac{b - x^2}{a\,t}$, P is power, x is distance and t is time. The dimensions of a and b are:

A) $[\mathrm{M\,L^2T^{-3}}]$, $[\mathrm{L}]$
B) $[\mathrm{M^{-1}L^2T^3}]$, $[\mathrm{L^2}]$
C) $[\mathrm{M^{-1}L^0T^2}]$, $[\mathrm{M^0L^2T^0}]$
D) $[\mathrm{M\,L^0T^{-2}}]$, $[\mathrm{L^2T}]$

7 Hawassa University · Test

Which of the following is dimensionally DIFFERENT from the others?

A) Work
B) Kinetic energy
C) Angular velocity
D) Torque

8 Jimma University · Model Exam

The speed limit in an Ethiopian city is 30 km/h. What is this speed in miles per hour? (1 mile = 1.609 km)

A) 48.3 mi/h
B) 30 mi/h
C) 8.3 mi/h
D) 18.6 mi/h

9 Jimma University · Model Exam

A house cat's average body temperature is 101.5 °F. What is this in degrees Celsius?

A) 38.6 °C
B) 56.4 °C
C) 69.5 °C
D) 74.2 °C

10 Haramaya University · Model Exam

The period of a simple pendulum is suspected to be $T = k\,l^a g^b$, where l is the length, g is the acceleration due to gravity and k is a dimensionless constant. Dimensional analysis gives:

A) $a = 1,\ b = -1$
B) $a = \frac{1}{2},\ b = \frac{1}{2}$
C) $a = -\frac{1}{2},\ b = \frac{1}{2}$
D) $a = \frac{1}{2},\ b = -\frac{1}{2}$

Answers & Explanations

1. **C**

N/m²

Density $= m/V$, so its units are mass per volume. N/m² (newton per square metre) is the pascal, the unit of **pressure**.

2. **C**

Amount of substance

Amount of substance (mole) is one of the 7 SI base quantities. Area $= l \times w$, power $= W/t$ and density $= m/V$ are all derived.

3. **False**

Force is **derived**: $F = ma$, so its unit is $\mathrm{kg \cdot m/s^2}$ (newton) and its dimension is $\mathrm{MLT^{-2}}$.

4. **A**

3.54×10^{-3} **μm**

Convert all to metres:
A) $3.54 \times 10^{-3} \times 10^{-6} = 3.54 \times 10^{-9}$ m
B) $3.54 \times 10^{5} \times 10^{-3} = 354$ m
C) $3.54 \times 10^{-3} \times 10^{-3} = 3.54 \times 10^{-6}$ m
D) $3.54 \times 10^{4} \times 10^{-6} = 3.54 \times 10^{-2}$ m
The smallest is A.

5. **27.8 m/s; [LT^{-1}]**

$100 \times \dfrac{1000\ \mathrm{m}}{3600\ \mathrm{s}} = 27.8$ m/s. Velocity = displacement/time, so $[v] = \mathrm{L/T} = \mathrm{LT^{-1}}$.

6. **C**

$[\mathrm{M^{-1}L^{0}T^{2}}]$, $[\mathrm{M^{0}L^{2}T^{0}}]$

b is subtracted from x^2, so $[b] = \mathrm{L^2}$. Then $[a] = [b]/([P][t]) = \mathrm{L^2/(ML^2T^{-3} \cdot T)} = \mathrm{M^{-1}T^{2}}$.

7. **C**

Angular velocity

Work, kinetic energy and torque all have dimension $\mathrm{ML^2T^{-2}}$. Angular velocity (rad/s) has dimension $\mathrm{T^{-1}}$.

8. **D**

18.6 mi/h

$30\ \mathrm{km/h} \times \dfrac{1\ \mathrm{mile}}{1.609\ \mathrm{km}} = 18.6$ mi/h. (This is the module's own activity question.)

9. **A**

38.6 °C

$T_C = \frac{5}{9}(101.5 - 32) = \frac{5}{9}(69.5) = 38.6$ °C.

10. **D**

$a = \frac{1}{2},\ b = -\frac{1}{2}$

$[T] = \mathrm{T} = \mathrm{L}^a(\mathrm{LT}^{-2})^b = \mathrm{L}^{a+b}\mathrm{T}^{-2b}$.

Time: $-2b = 1 \Rightarrow b = -\frac{1}{2}$. Length: $a + b = 0 \Rightarrow a = \frac{1}{2}$.

So $T = k\sqrt{l/g}$ (the real formula has $k = 2\pi$).

CHAPTER 1 · LESSON 1.2

Uncertainty in Measurement and Significant Figures

AFTER THIS LESSON YOU CAN

- Distinguish systematic and random errors
- Distinguish accuracy, precision and calibration
- State the uncertainty of scale and digital instruments
- Count significant figures and apply the rules for calculations

1.2.1 Why every measurement is uncertain

No measurement is perfectly exact. Every instrument has a limit, and every observer makes small mistakes. The **uncertainty** gives the **range of possible values** that is believed to contain the true value. It tells us **how much we can trust** a measurement. (For historical reasons, uncertainty analysis is often called *error analysis*.)

D TWO KINDS OF ERROR

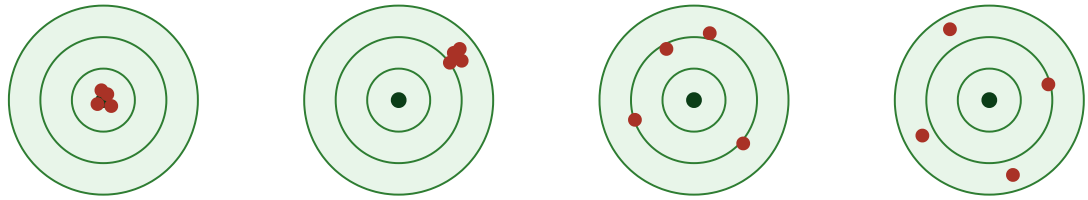
	Systematic error	Random error
Cause	Instrument **out of calibration** (faulty instrument, zero error)	Natural **fluctuation** of repeated readings; limited fineness of the scale
Effect	Readings **consistently too large or too small** (always in one direction)	Readings **equally likely to be too large or too small**, scattered around the average
Cure	**Eliminated** by **pre-calibrating** against a trusted standard	**Reduced** (not eliminated) by **taking many readings and averaging**

! CALIBRATION REMOVES SYSTEMATIC ERRORS ONLY

"Random errors can be eliminated by pre-calibrating against a known standard" → **False**. Calibration fixes **systematic** errors. Random errors can only be **reduced by averaging** many readings.

Accuracy, precision, calibration and sensitivity

Term	Meaning
Accuracy	How close a measurement is to the **true (accepted) value**
Precision	How close **repeated measurements are to each other** (the reproducibility of results)
Calibration	**Checking an instrument's scale against an accepted standard**, then fixing and dividing the scale to ensure accurate readings
Sensitivity	The **response of an instrument to a unit change of input** (this is *not* calibration)



Accurate + precise — **Precise, not accurate** (systematic error) — **Accurate on average,** not precise (random error) — **Neither**

Accuracy = close to the bull's-eye (true value); precision = close to each other

1.2.2 Uncertainty of a single reading

F RULES OF THUMB FOR A SINGLE MEASUREMENT

- **Scale (analog) device** (metre stick, thermometer): uncertainty = **½ × smallest division**. A metre stick with 1 mm divisions has an uncertainty of **±0.05 cm**.
- **Digital device** (digital balance, stopwatch): uncertainty = **the smallest increment** shown. A balance reading 5.7513 kg has an uncertainty of **±0.0001 kg**.
- **Reporting:** measurement $= x_{\text{best}} \pm \Delta x$, e.g. $L = 5.7 \pm 0.05$ cm, meaning $5.65\ \text{cm} \le L \le 5.75\ \text{cm}$.

Repeated readings: mean, absolute error, relative and percentage error

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}, \qquad \Delta x_i = |x_i - \bar{x}|, \qquad \overline{\Delta x} = \frac{\sum |x_i - \bar{x}|}{n}$$

$$\text{relative error} = \frac{\overline{\Delta x}}{\bar{x}}, \qquad \text{percentage error} = \frac{\overline{\Delta x}}{\bar{x}} \times 100\%$$

✎ WORKED EXAMPLE 1: PENDULUM TIMINGS (AAU MID EXAM)

Five measurements of a pendulum's period: 2.63, 2.56, 2.42, 2.71, 2.80 s. Find the mean, the mean absolute error and the percentage error.

Mean: $\bar{T} = \dfrac{13.12}{5} = 2.624$ s.

Absolute errors: 0.006, 0.064, 0.204, 0.086, 0.176 s. Mean absolute error: $\dfrac{0.536}{5} = 0.107 \approx 0.11$ s.

Relative error: $\dfrac{0.107}{2.624} = 0.041$, so the percentage error ≈ **4.1%**.

Result: $T = 2.62 \pm 0.11$ s.

Propagation of uncertainty (how errors combine)

COMBINING UNCERTAINTIES

Operation	Rule	Example
Sum or difference $A \pm B$	**Add absolute** uncertainties: $\Delta R = \Delta A + \Delta B$	$(3.56 \pm 0.05) + (3.25 \pm 0.04) = 6.81 \pm 0.09$
Product or quotient AB, A/B	**Add relative (percentage)** uncertainties: $\frac{\Delta R}{R} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$	$R = V/I$ with 5% and 2% errors → **7%**
Power A^n	Multiply the relative uncertainty by n: $\frac{\Delta R}{R} = n\frac{\Delta A}{A}$	the area of a square with 1% error in its side → **2%**

1.2.3 Significant figures

The **significant figures** (significant digits) of a number are the digits that are **known reliably plus one estimated digit**. Writing 0.428 m means you know the length to about 0.001 m. Significant figures have **nothing to do with the number of decimal places**: 0.428 m = 42.8 cm, three significant figures either way.

COUNTING SIGNIFICANT FIGURES: THE RULES

1. **All non-zero digits** are significant: 4.62 → 3.
2. **Zeros between** non-zero digits are significant: 1005 → 4; 1.0005 → 5.
3. **Leading zeros** (that only locate the decimal point) are **not** significant: 0.0062 → 2; 0.000098 → 2.
4. **Trailing zeros after a decimal point** are significant: 4.00 → 3; 4.0500 → 5.
5. **Trailing zeros in a whole number** are ambiguous: write 300 as 3×10^2 (1 s.f.) or 3.00×10^2 (3 s.f.).
6. In **scientific notation**, count the digits of the coefficient: 1.118×10^{-6} → 4.

Short rule (module): a zero is significant if a **non-zero digit is anywhere to its left**.

"ZEROS THAT FOLLOW A NUMBER ARE ALWAYS SIGNIFICANT" → FALSE

Trailing zeros are significant only when they follow a decimal point (3.00) or when stated by scientific notation. In "300" they are not necessarily significant (Ambo exam).

Rules for calculations

THE TWO CALCULATION RULES

Rule 1: multiplication and division. The answer has the same number of **significant figures** as the factor with the **fewest significant figures**.

Rule 2: addition and subtraction. The answer has the same number of **decimal places** as the term with the **fewest decimal places**.

WORKED EXAMPLE 2: APPLYING THE RULES

- $8.71\ \text{cm} \times 3.2\ \text{cm} = 27.872 \rightarrow$ **28 cm²** (3.2 has 2 s.f.)
- $9.65 + 8.4 - 2.89 = 15.16 \rightarrow$ **15.2 cm** (8.4 has 1 decimal place)
- $436.32 + 227.2 + 0.301 = 663.821 \rightarrow$ **663.8**
- $9.99\ \text{m} + 0.0099\ \text{m} = 9.9999 \rightarrow$ **10.00 m** (2 decimal places)
- $0.25 \times 0.1098 = 0.02745 \rightarrow$ **0.027** (2 s.f.)

ROUND ONLY AT THE END

Keep extra digits in intermediate steps and round **only the final answer**. Rounding at each step makes errors accumulate.

Exam Practice

18 QUESTIONS · LESSON 1.2

1 University of Gondar · Mid Exam

An error caused by a faulty instrument is a:

A) Random error
B) Personal error
C) Parallax error
D) Systematic error

2 Arba Minch University · Mid Exam

Fill in the blank: "An error resulting from measuring devices being out of calibration is called _____."

3 Wolaita Sodo University · Mid Exam

Random errors can be eliminated by pre-calibrating against a known, trusted standard.

True / False

4 Addis Ababa University · Final 2021

Which of the following statements is NOT correct about calibration?

A) Checking the scale of the instrument against an accepted standard to ensure accuracy of readings
B) Response of the instrument to a unit change of input
C) Dividing the scale into an appropriate number of intervals/graduation marks
D) Fixing or setting the scale to ensure the accuracy of absolute values

5 Ambo University · Mid Exam

Zeros that follow a number are always significant figures.

True / False

6 Arba Minch University · Mid Exam

Which of the following statements about significant figures is WRONG?

A) All non-zero numbers are significant
B) Zeros within a number are always significant
C) Zeros that follow a number may be significant
D) Zeros that do nothing but set the decimal point are significant

7 Haramaya University · Mid Exam

How many significant figures does 1.118×10^{-6} V have?

A) 3
B) 4
C) 6
D) 10

8 Ambo University · Mid Exam

The product 2.5×1.5 written to the correct number of significant figures is:

A) 3.75
B) 3.7
C) 3.8
D) 4

9 Dilla University · Mid Exam 2012

$0.25 \times 0.1098 = ?$

A) 0.02745
B) 0.027
C) 0.0275
D) 0.03

10 Arba Minch University · Mid Exam

A rectangle measures 12.71 m by 3.46 m. Its area to the correct significant figures is:

A) 43.9766 m²
B) 44.0 m²
C) 44 m²
D) 43.98 m²

11 Arba Minch University · Mid Exam

The sides of a plate are 3.7 cm and 4.85 cm. Its area is:

A) 17.945 cm²
B) 17.9 cm²
C) 18 cm²
D) 17.95 cm²

12 University of Gondar · Mid Exam

$436.32 + 227.2 + 0.301 = ?$

A) 663.821
B) 663.82
C) 663.8
D) 664

13 Hawassa University · Test

$9.99\ \text{m} + 0.0099\ \text{m}$ correctly reported is:

A) 10.00 m
B) 10.0 m
C) 9.9999 m
D) 10 m

14 University of Gondar · Mid Exam

If $A = 3.56 \pm 0.05$ and $B = 3.25 \pm 0.04$, then:

A) $6.77 \le A + B \le 6.85$
B) $6.72 \le A + B \le 6.90$
C) $6.81 \le A + B \le 6.90$
D) $6.76 \le A + B \le 6.86$

15 Hawassa University · Test

$V = 100 \pm 5$ V and $I = 10 \pm 0.2$ A. The percentage error in $R = V/I$ is:

A) 3%
B) 7%
C) 2.5%
D) 10%

16 Arsi University · Mid Exam 2014

If $L = 5.0 \pm 0.01$ m, the percentage uncertainty is:

A) 0.20%
B) 2.0%
C) 0.01%
D) 5.0%

17 Addis Ababa University · Mid Exam

Five measurements of a pendulum's period are 2.63, 2.56, 2.42, 2.71 and 2.80 s. Find (i) the mean, (ii) the mean absolute error, and (iii) the percentage error.

18 Addis Ababa University · Final 2021

Define precision and say how it differs from accuracy.

Answers & Explanations

1. D

Systematic error

A faulty (badly calibrated) instrument shifts every reading in the same direction, which is a systematic error. It is removed by calibration.

2. Systematic error

The module defines systematic errors as errors resulting from measuring devices being out of calibration.

3. False

Calibration eliminates **systematic** errors. Random errors are reduced by repeating measurements and averaging.

4. B

Response of the instrument to a unit change of input

"Response of the instrument to a unit change of input" defines **sensitivity**, not calibration.

5. False

In 300 the trailing zeros are not necessarily significant (write 3×10^2). In 3.00 they are significant, because they follow the decimal point.

6. D

Zeros that do nothing but set the decimal point are significant

Zeros that only locate the decimal point (as in 0.0005) are **not** significant.

7. B

4

In scientific notation, count the digits of the coefficient: 1, 1, 1, 8 → 4 significant figures.

8. C

3.8

Both factors have 2 s.f., so the answer must have 2 s.f.: $3.75 \rightarrow 3.8$.

9. B

0.027

0.25 has 2 s.f. (leading zeros don't count), so the answer needs 2 s.f.: $0.02745 \rightarrow 0.027$.

10. B

44.0 m²

$12.71 \times 3.46 = 43.9766$. 3.46 has 3 s.f., so round to 3 s.f.: **44.0 m²**. (The zero is significant and must be written.)

11. C

18 cm²

3.7 has only 2 s.f., so $17.945 \rightarrow 18$ cm².

12. C

663.8

For addition, keep the fewest **decimal places**. 227.2 has 1 decimal place, so $663.821 \rightarrow 663.8$.

13. A

10.00 m

9.99 has 2 decimal places, so $9.9999 \rightarrow 10.00$ m (two decimal places).

14. B

$6.72 \leq A + B \leq 6.90$

For a sum, add absolute uncertainties: $A + B = 6.81 \pm 0.09$, so $6.72 \leq A + B \leq 6.90$.

15. **B**

7%

For a quotient, add percentage errors: $\frac{5}{100} = 5\%$ and $\frac{0.2}{10} = 2\%$, so the total is **7%**.

16. **A**

0.20%

$\frac{0.01}{5.0} \times 100\% = 0.20\%$.

17. **2.624 s; 0.11 s; 4.1%**

(i) $\bar{T} = 13.12/5 = 2.624$ s.

(ii) Deviations: 0.006, 0.064, 0.204, 0.086, 0.176 → mean $= 0.536/5 = 0.107 \approx 0.11$ s.

(iii) $0.107/2.624 \times 100\% = 4.1\%$.

18. **Precision = closeness of repeated measurements to each other**

Accuracy is closeness to the true value. A badly calibrated instrument can be very **precise** (consistent readings) but **inaccurate** (all readings shifted).

CHAPTER 1 · LESSON 1.3

Vectors: Composition and Resolution

AFTER THIS LESSON YOU CAN

- Distinguish scalars and vectors
- Add vectors graphically and by the parallelogram law
- Resolve vectors into components and use unit vectors
- Use the dot and cross products

1.3.1 Scalars and vectors

	Scalar	Vector
Specified by	**Magnitude** (number + unit) only	**Magnitude AND direction**
Algebra	Ordinary algebra	Vector algebra
Examples	mass, time, volume, **speed**, distance, energy, temperature, work	**displacement**, **velocity**, acceleration, force, momentum, weight

Representation. Algebraically, a vector is written with an arrow ($\vec{A}$) or in bold (**A**). Its **magnitude** is $|\vec{A}|$ or A, always positive. Geometrically, it is an **arrow**: its **length** shows the magnitude and its **head** shows the direction.

★ **BASIC VECTOR FACTS**

- A vector **changes** if its magnitude, its direction, or **both** change.
- Vectors can be **moved parallel to themselves** without changing them.
- Multiplying by a positive number changes only the **magnitude**; a **negative** number also **reverses the direction**. $-\vec{A}$ has the same magnitude as $\vec{A}$ but the opposite direction.
- Multiplying a vector by a **scalar with units** gives a **new physical quantity** (velocity × time = displacement).

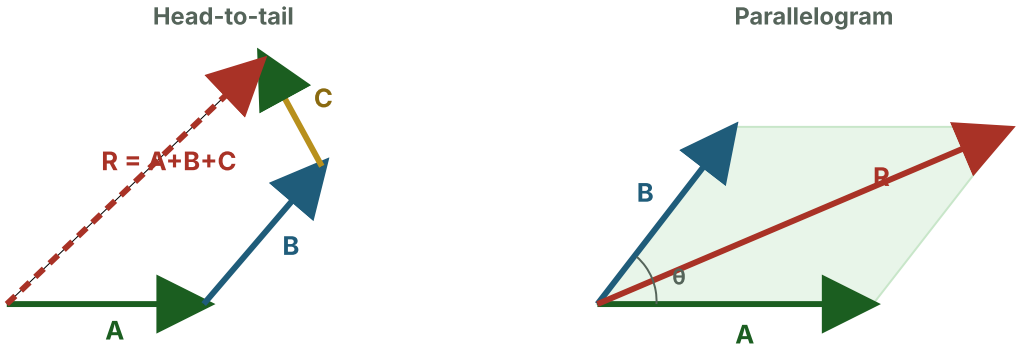
1.3.2 Adding vectors

A. Graphical (head-to-tail, polygon) method

Place the **tail of each vector at the head of the previous one**. The **resultant** $\vec{R}$ runs from the **tail of the first** to the **head of the last**. The order does not matter: $\vec{A} + \vec{B} = \vec{B} + \vec{A}$.

B. Parallelogram law

Place $\vec{A}$ and $\vec{B}$ **tail to tail** as adjacent sides of a parallelogram. The **diagonal** from the common tail is the resultant.



Both methods give the same resultant

F **MAGNITUDE AND DIRECTION OF THE RESULTANT OF TWO VECTORS**

If θ is the angle **between** $\vec{A}$ and $\vec{B}$ (tail to tail):

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta} \quad \text{(cosine law)}$$

The angle α between $\vec{R}$ and $\vec{A}$ follows from the sine law: $\dfrac{B}{\sin\alpha} = \dfrac{R}{\sin\theta}$.

Special cases: $\theta = 0^\circ$ gives $R_{\max} = A + B$; $\theta = 180^\circ$ gives $R_{\min} = |A - B|$; $\theta = 90^\circ$ gives $R = \sqrt{A^2 + B^2}$.

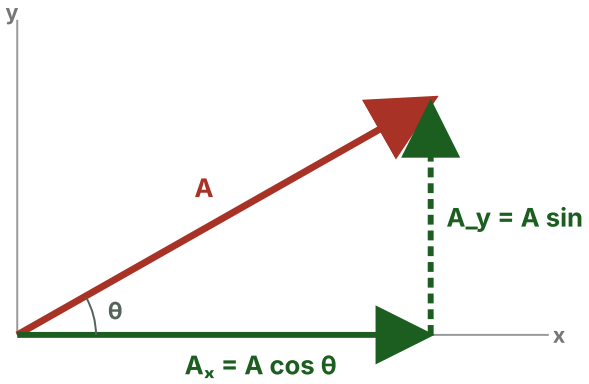
So always: $|A - B| \leq R \leq A + B$.

! **THE RESULTANT RANGE QUESTION**

Two vectors of magnitudes 5 and 6 can give a resultant **anywhere from 1 to 11**, so 12 is impossible. Two vectors of 12 and 8 give $R_{\max} = 20$ and $R_{\min} = 4$. And if $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, the vectors must be **perpendicular** ($\cos\theta = 0$).

1.3.3 Components of a vector

Any vector can be **resolved** into perpendicular components:



F **COMPONENTS (θ MEASURED FROM THE +X AXIS)**

$$A_x = A\cos\theta, \qquad A_y = A\sin\theta, \qquad A = \sqrt{A_x^2 + A_y^2}, \qquad \tan\theta = \frac{A_y}{A_x}$$

In 3D: $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$, with **direction cosines** $\cos\alpha = \frac{A_x}{A}$, $\cos\beta = \frac{A_y}{A}$, $\cos\gamma = \frac{A_z}{A}$, where $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

! **CHOOSE THE RIGHT QUADRANT**

$\tan^{-1}(A_y/A_x)$ on a calculator gives an angle between −90° and 90°. If $A_x < 0$, **add 180°**. Draw a sketch first. Also note that a component can **never be larger** than the vector itself.

1.3.4 Unit vectors

D **UNIT VECTOR**

A vector with **magnitude 1**, **dimensionless**, whose only job is to **point in a direction**. It is written with a hat: $\hat{A}$. The Cartesian unit vectors are $\hat{i}$ (+x), $\hat{j}$ (+y) and $\hat{k}$ (+z).

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}, \qquad \hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{A_x\hat{i} + A_y\hat{j} + A_z\hat{k}}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

Adding in unit-vector notation: add the matching components:

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$$

WORKED EXAMPLE 1: UNIT VECTOR OF A RESULTANT (ARBA MINCH EXAM)

$\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{B} = \hat{i} + \hat{j} + 2\hat{k}$, $\vec{C} = 3\hat{i} - 2\hat{j} + 3\hat{k}$. **Find the unit vector along** $\vec{R} = \vec{A} + \vec{B} + \vec{C}$.

$\vec{R} = (2 + 1 + 3)\hat{i} + (-3 + 1 - 2)\hat{j} + (1 + 2 + 3)\hat{k} = 6\hat{i} - 4\hat{j} + 6\hat{k}$

$|\vec{R}| = \sqrt{36 + 16 + 36} = \sqrt{88} = 2\sqrt{22}$

$$\hat{R} = \frac{6\hat{i} - 4\hat{j} + 6\hat{k}}{2\sqrt{22}} = \frac{3\hat{i} - 2\hat{j} + 3\hat{k}}{\sqrt{22}}$$

WORKED EXAMPLE 2: SOLVING A VECTOR EQUATION (ARBA MINCH EXAM)

$\vec{A} = 6\hat{i} - 8\hat{j}$, $\vec{B} = -8\hat{i} + 3\hat{j}$, $\vec{C} = 26\hat{i} + 19\hat{j}$. **Find** a **and** b **so that** $a\vec{A} + b\vec{B} + \vec{C} = 0$.

x-components: $6a - 8b + 26 = 0$. y-components: $-8a + 3b + 19 = 0$.

Solving: multiply the first equation by 3 and the second by 8, then add: $-46a + 230 = 0 \Rightarrow a = 5$.

Then $b = (6a + 26)/8 = 56/8 = 7$.

Answer: $a = 5,\ b = 7$.

1.3.5 Products of vectors (frequently examined)

DOT (SCALAR) PRODUCT

$$\vec{A}\cdot\vec{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$$

- The result is a **scalar**.
- $\hat{i}\cdot\hat{i} = 1,\ \hat{i}\cdot\hat{j} = 0.$
- **Perpendicular** $\Leftrightarrow \vec{A}\cdot\vec{B} = 0.$
- Angle: $\cos\theta = \dfrac{\vec{A}\cdot\vec{B}}{AB}.$
- Component of $\vec{A}$ along $\vec{B}$: $\dfrac{\vec{A}\cdot\vec{B}}{B}.$
- Physics example: work $W = \vec{F}\cdot\vec{d}.$

CROSS (VECTOR) PRODUCT

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

- The result is a **vector perpendicular to both** $\vec{A}$ and $\vec{B}$ (right-hand rule).
- Magnitude $AB\sin\theta$ = the **area of the parallelogram** formed by $\vec{A}$ and $\vec{B}$.
- $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A};\ \hat{i}\times\hat{j} = \hat{k}.$
- **Parallel** $\Leftrightarrow \vec{A}\times\vec{B} = 0.$
- Physics example: torque $\vec{\tau} = \vec{r}\times\vec{F}.$

WORKED EXAMPLE 3: DOT AND CROSS PRODUCTS (AASTU MID EXAM)

$\vec{A} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{B} = -3\hat{i} - \hat{j} + 4\hat{k}$. Find (a) $\vec{A}\cdot\vec{B}$; (b) a vector perpendicular to both; (c) a vector of magnitude 4 perpendicular to both.

(a) $\vec{A}\cdot\vec{B} = (1)(-3) + (2)(-1) + (-2)(4) = -13.$

(b) $\vec{A}\times\vec{B} = \hat{i}(2\cdot 4 - (-2)(-1)) - \hat{j}(1\cdot 4 - (-2)(-3)) + \hat{k}(1\cdot(-1) - 2\cdot(-3)) = 6\hat{i} + 2\hat{j} + 5\hat{k}.$

(c) $|\vec{A}\times\vec{B}| = \sqrt{36+4+25} = \sqrt{65}$, so $\vec{V} = \dfrac{4}{\sqrt{65}}(6\hat{i} + 2\hat{j} + 5\hat{k}).$

$(\vec{A}\times\vec{B})\cdot\vec{A} = 0$ ALWAYS

$\vec{A}\times\vec{B}$ is perpendicular to $\vec{A}$, so its dot product with $\vec{A}$ is **zero**, whatever the angle between $\vec{A}$ and $\vec{B}$ (Hawassa test).

Exam Practice

18 QUESTIONS · LESSON 1.3

1 Ambo University · Mid Exam

Two vectors have magnitudes 12 units and 8 units. The maximum and minimum magnitudes of their resultant are:

A) 96 units, 1.5 units
B) 20 units, 0
C) 20 units, 4 units
D) 14.4 units, 4 units

2 Arba Minch University · Mid Exam

$|\vec{A}| = 5$ units and $|\vec{B}| = 6$ units. Which of the following CANNOT be the magnitude of $\vec{A} + \vec{B}$?

A) 10 units
B) 12 units
C) 5 units
D) 1 unit

3 Wachemo University · Mid Exam 2013

If $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$, then $\vec{A}$ and $\vec{B}$ are:

A) Perpendicular to each other
B) Parallel
C) Antiparallel
D) Equal

4 Haramaya University · Mid Exam

Two vectors of equal magnitude 5.00 have a resultant of magnitude 6.00. The angle between them is about:

A) 106°
B) 74°
C) 53°
D) 37°

5 Haramaya University · Mid Exam

The unit vector along $\vec{A} = 3\hat{i} - 4\hat{j}$ is:

A) $3\hat{i} - 4\hat{j}$
B) $0.8\hat{i} - 0.6\hat{j}$
C) $0.6\hat{i} - 0.8\hat{j}$
D) $\frac{1}{5}(4\hat{i} - 3\hat{j})$

6 Wolaita Sodo University · Final Exam 2014

Which of the following is NOT a unit vector?

A) $\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$
B) $\hat{i}+\hat{j}$
C) $\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$
D) $\frac{1}{\sqrt{6}}(\hat{i}+2\hat{j}+\hat{k})$

7 AASTU · Mid Exam 2

If $\vec{A} = (2\text{ m})\hat{i} - (3\text{ m})\hat{j}$ and $\vec{B} = (1\text{ m})\hat{i} - (2\text{ m})\hat{j}$, then $\vec{A} - 2\vec{B} =$

A) $(1\text{ m})\hat{j}$
B) $(-1\text{ m})\hat{j}$
C) $(4\text{ m})\hat{i} - (7\text{ m})\hat{j}$
D) $(4\text{ m})\hat{i} + (1\text{ m})\hat{j}$

8 Arba Minch University · Mid Exam

$\vec{A} = 2\hat{i} + \hat{j} + \hat{k}$ is perpendicular to $\vec{B} = \hat{i} - 4\hat{j} + \lambda\hat{k}$. The value of λ is:

A) -2
B) 0
C) 2
D) 4

9 Arba Minch University · Mid Exam

$\vec{A} = 6\hat{i} - 8\hat{j}$, $\vec{B} = -8\hat{i} + 3\hat{j}$, $\vec{C} = 26\hat{i} + 19\hat{j}$ and $a\vec{A} + b\vec{B} + \vec{C} = 0$. Then:

A) $a = 7,\ b = 5$
B) $a = 5,\ b = 7$
C) $a = -5,\ b = 7$
D) $a = 5,\ b = -7$

10 Bahir Dar University · Mid Exam

$\vec{A} = 3\hat{i} - 3\hat{j}$, $\vec{B} = -\hat{i} - 4\hat{j}$, $\vec{C} = -5\hat{i} - 5\hat{j}$ and $r\vec{A} - k\vec{B} + \vec{C} = 0$. Then:

A) $r = 2,\ k = 1$
B) $r = -1,\ k = 2$
C) $r = 1,\ k = 2$
D) $r = 1,\ k = -2$

11 Hawassa University · Test

If θ is the angle between $\vec{A}$ and $\vec{B}$, then $(\vec{A} \times \vec{B}) \cdot \vec{A}$ equals:

A) $A^2B\sin\theta$ **B)** $A^2B\cos\theta$

C) $AB\sin\theta\cos\theta$ **D)** Zero

12 University of Gondar · Mid Exam

Given $\vec{A} + \vec{B} = 6\hat{i} + \hat{j}$ and $\vec{A} - \vec{B} = -4\hat{i} + 7\hat{j}$, find $\vec{A}$ and $\vec{B}$.

13 Ambo University · Mid Exam

$\vec{s}_1 = (3\hat{i} - 4\hat{j} + 4\hat{k})$ m and $\vec{s}_2 = (2\hat{i} + 3\hat{j} - \hat{k})$ m. Find (a) $\vec{s}_1 + \vec{s}_2$, (b) $3\vec{s}_1 - 2\vec{s}_2$, (c) $|\vec{s}_1 + \vec{s}_2|$.

14 Bahir Dar University · Mid Exam

$\vec{A} = 3\hat{i} - 4\hat{j} + 4\hat{k}$ and $\vec{B} = 2\hat{i} - 3\hat{j} + \hat{k}$. Find the unit vector in the direction of $2\vec{B} - \vec{A}$.

15 Adama Science and Technology University · Exam

$\vec{A} = 2\hat{i} - 3\hat{j}$ and $\vec{B} = -\hat{i} + 8\hat{k}$. Find (a) the angle between them and (b) the projection of $\vec{A}$ along $\vec{B}$.

16 Adama Science and Technology University · Exam

Find a unit vector normal to the plane containing $\vec{A} = 4\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = 2\hat{i} - 3\hat{j} + 2\hat{k}$.

17 Hawassa University · Test

For what values of a are $\vec{P} = a\hat{i} + a\hat{j} + 3\hat{k}$ and $\vec{Q} = a\hat{i} - 2\hat{j} - \hat{k}$ perpendicular?

18 Wolaita Sodo University · Final Exam 2014

The component of a vector is always larger than the magnitude of the vector.

True / False

Answers & Explanations

1. C

20 units, 4 units

$R_{\max} = A + B = 20$ (same direction); $R_{\min} = |A - B| = 4$ (opposite directions).

2. B

12 units

$|A - B| \le R \le A + B$, so $1 \le R \le 11$. Twelve is outside the range.

3. **A**

Perpendicular to each other

Squaring: $A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 - 2AB\cos\theta \Rightarrow \cos\theta = 0 \Rightarrow \theta = 90^\circ$.

4. **A**

106°

$36 = 25 + 25 + 50\cos\theta \Rightarrow \cos\theta = -0.28 \Rightarrow \theta \approx 106^\circ$.

5. **C**

$0.6\hat{i} - 0.8\hat{j}$

$|\vec{A}| = \sqrt{9+16} = 5$, so $\hat{A} = (3\hat{i} - 4\hat{j})/5 = 0.6\hat{i} - 0.8\hat{j}$.

6. **B**

$\hat{i} + \hat{j}$

$|\hat{i} + \hat{j}| = \sqrt{2} \neq 1$. The others have magnitude 1 (check: $\sin^2\theta + \cos^2\theta = 1$; $(1+4+1)/6 = 1$).

7. **A**

$(1\text{ m})\hat{j}$

$2\vec{B} = 2\hat{i} - 4\hat{j}$, so $\vec{A} - 2\vec{B} = (2-2)\hat{i} + (-3+4)\hat{j} = (1\text{ m})\hat{j}$.

8. **C**

2

$\vec{A} \cdot \vec{B} = 2 - 4 + \lambda = 0 \Rightarrow \lambda = 2$.

9. **B**

$a = 5,\ b = 7$

x: $6a - 8b + 26 = 0$; y: $-8a + 3b + 19 = 0$. Solving gives $a = 5,\ b = 7$ (see Worked example 2).

10. **C**

$r = 1,\ k = 2$

x: $3r + k - 5 = 0$; y: $-3r + 4k - 5 = 0$. Adding: $5k - 10 = 0 \Rightarrow k = 2$, then $r = 1$.

11. **D**

Zero

$\vec{A} \times \vec{B}$ is perpendicular to $\vec{A}$, so their dot product is zero.

12. $\vec{A} = \hat{i} + 4\hat{j},\ \vec{B} = 5\hat{i} - 3\hat{j}$

Adding the equations: $2\vec{A} = 2\hat{i} + 8\hat{j} \Rightarrow \vec{A} = \hat{i} + 4\hat{j}$.
Then $\vec{B} = (6\hat{i} + \hat{j}) - \vec{A} = 5\hat{i} - 3\hat{j}$.

13. **(5i − j + 3k) m; (5i − 18j + 14k) m; √35 m**

(a) Add components: $(3+2)\hat{i} + (-4+3)\hat{j} + (4-1)\hat{k} = 5\hat{i} - \hat{j} + 3\hat{k}$.
(b) $(9\hat{i} - 12\hat{j} + 12\hat{k}) - (4\hat{i} + 6\hat{j} - 2\hat{k}) = 5\hat{i} - 18\hat{j} + 14\hat{k}$.
(c) $\sqrt{25 + 1 + 9} = \sqrt{35} \approx 5.92$ m.

14. $(\hat{i} - 2\hat{j} - 2\hat{k})/3$

$2\vec{B} - \vec{A} = (4-3)\hat{i} + (-6+4)\hat{j} + (2-4)\hat{k} = \hat{i} - 2\hat{j} - 2\hat{k}$.
Its magnitude is $\sqrt{1+4+4} = 3$, so the unit vector is $(\hat{i} - 2\hat{j} - 2\hat{k})/3$.

15. ≈ 93.9°; −2/√65

$\vec{A} \cdot \vec{B} = (2)(-1) + 0 + 0 = -2$; $A = \sqrt{13}$, $B = \sqrt{65}$.
(a) $\cos\theta = -2/\sqrt{845} = -0.0688 \Rightarrow \theta \approx 93.9^\circ$.
(b) Projection $= \vec{A} \cdot \vec{B}/B = -2/\sqrt{65} \approx -0.25$.

16. $(3\hat{i} - 10\hat{j} - 18\hat{k})/\sqrt{433}$

The normal is $\vec{A} \times \vec{B} = \hat{i}(6-3) - \hat{j}(8+2) + \hat{k}(-12-6) = 3\hat{i} - 10\hat{j} - 18\hat{k}$.
Its magnitude is $\sqrt{9+100+324} = \sqrt{433}$, so the unit normal is $(3\hat{i} - 10\hat{j} - 18\hat{k})/\sqrt{433}$ (or its negative).

17. a = 3 or a = −1

$\vec{P} \cdot \vec{Q} = a^2 - 2a - 3 = 0 \Rightarrow (a-3)(a+1) = 0$.

18. False

$|A_x| = A|\cos\theta| \leq A$. A component can never exceed the magnitude of the vector.

CHAPTER 1 · QUICK REVISION

Preliminaries — at a glance

Formula sheet: Chapter 1

Topic	Formula / rule
Speed conversion	km/h ÷ 3.6 = m/s
Area / volume conversion	1 m² = 10⁴ cm²; 1 m³ = 10⁶ cm³ = 1000 L
Temperature	$T_C = \frac{5}{9}(T_F - 32)$; $T_K = T_C + 273.15$
Scale device uncertainty	½ × smallest division
Digital device uncertainty	smallest increment
Sum or difference	add **absolute** uncertainties
Product or quotient	add **relative (%)** uncertainties
× and ÷	fewest **significant figures**
+ and −	fewest **decimal places**
Resultant of two vectors	$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$; $\lvert A - B\rvert \le R \le A + B$
Components	$A_x = A\cos\theta$, $A_y = A\sin\theta$, $\tan\theta = A_y/A_x$
Unit vector	$\hat{A} = \vec{A}/\lvert\vec{A}\rvert$
Dot product	$AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$; perpendicular ⇔ 0
Cross product	magnitude $AB\sin\theta$; perpendicular to both; parallel ⇔ 0

CHAPTER 1 MISTAKES TO AVOID

1. N/m² is pressure, not density; mole is basic; force is derived.
2. Calibration removes **systematic** errors; averaging reduces **random** errors.
3. Leading zeros are never significant; trailing zeros after a decimal point always are.
4. × ÷ → significant figures; + − → decimal places.
5. Square or cube conversion factors for areas and volumes.
6. Use the angle **between** the vectors (tail to tail) in the cosine law.